

Review Session 4

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1 Lebesgue Decomposition Theorem

Let $(X, \mathbb{X})$ be a measure space. Let λ, μ be measures on a σ -field $\mathbb{X}$. We call λ *singular* w.r.t. μ and denote

$$\lambda \perp \mu$$

if and only if there exists sets $A, B \in \mathbb{X}$ such that

$$\lambda(A) = \mu(B) = 0$$

where $X = A \cup B, A \cap B = \emptyset$. We call λ absolutely continuous w.r.t. μ and denote

$$\lambda \ll \mu$$

if and only if

$$\mu(A) = 0 \implies \lambda(A) = 0$$

for all $A \in \mathbb{X}$.

Theorem 1 (Lebesgue Decomposition Theorem). *Let λ, μ be σ -finite measures on a σ -field $\mathbb{X}$. Then, there exists a unique decomposition λ_1, λ_2 of λ such that*

$$\lambda = \lambda_1 + \lambda_2$$

with

$$\lambda_1 \perp \mu, \lambda_2 \ll \mu$$

Corollary 1. *Let F, μ be spectral and Lebesgue measures respectively, each defined on measure space $([-\pi, \pi], \mathcal{B}([-\pi, \pi]))$. By Lebesgue decomposition theorem, there exists a unique decomposition of F such that*

$$F = F_1 + F_2$$

such that

$$F_1 \perp \mu, F_2 \ll \mu$$

This means that

1. There exists A, B with $A \cup B = [-\pi, \pi]$, $A \cap B = \emptyset$ such that

$$F_1(A) = F_2(B) = 0$$

2. For all $A \in \mathcal{B}([-\pi, \pi])$, $F_2(A) = 0$ if $\mu(A) = 0$.

Here, F_1 is associated with the singular, deterministic part of Wold decomposition. F_2 is associated with the nonsingular linear process part of Wold decomposition. The nonsingular part has a spectral density function because it is absolutely continuous w.r.t. Lebesgue measure. Let the Wold decomposition of X_t be given by

$$X_t = \sum_{k=0}^{\infty} \varphi_k \varepsilon_{t-k} + D_t$$

- Spectral distribution of $\sum_{k=0}^{\infty} \varphi_k \varepsilon_{t-k}$ will be absolutely continuous w.r.t. Lebesgue measure.
- Spectral distribution of D_t will not have Radon–Nikodym derivative.

2 OLS

Suppose we have a pair of random variables (x, y) , both of which are $I(0)$. For any β , we may always write

$$y = x'\beta + u$$

by defining $u := y - x'\beta$. Since both (x, y) are $I(0)$, u is also $I(0)$ for any choice of β . If we define u so that x and u is orthogonal, by projection theorem β is uniquely identified by $\beta = (\mathbb{E}[xx'])^{-1}\mathbb{E}[xy]$ as long as the inverse exists. Suppose we have data on (x, y) given by $(x_t, y_t)_{t=1}^n$ collected in data matrices,

$$Y = X\beta + U$$

Then, we may estimate β by

$$\hat{\beta} = \arg \min_{\beta} (Y - X\beta)'(Y - X\beta)$$

The FOC is given by

$$-2X'(Y - X\beta) = 0$$

so that we have

$$\hat{\beta} = (X'X)^{-1}X'Y$$

The variance of our estimate $\hat{\beta}$ is given by

$$\begin{aligned} \text{Var}(\hat{\beta}|X) &= \text{Var}(\beta + (X'X)^{-1}X'U|X) \\ &= (X'X)^{-1}X'\text{Var}(U|X)X(X'X)^{-1} \\ &= (X'X)^{-1}X'\Sigma X(X'X)^{-1} \end{aligned}$$

where we assume $\Sigma = \text{Var}(U|X)$, i.e., homoskedasticity. Furthermore, if Σ is spherical, i.e., $\Sigma = \sigma^2 I_n$, the OLS is BLUE. If we further assume that u is normally distributed, we have

$$\frac{\hat{\beta}_j}{\text{se}(\hat{\beta}_j)} \stackrel{d}{=} t_{n-p-1}$$

under $H_0 : \beta_j = 0$ where $\text{se}(\hat{\beta}_j)$ is square root of (j, j) -th element of $\widehat{\text{Var}(\hat{\beta}|X)} = \hat{\sigma}^2(X'X)^{-1}$. If we only assume $u \sim I(0)$, we have

$$\frac{\hat{\beta}_j}{\text{se}(\hat{\beta}_j)} \rightarrow_d \mathbb{N}(0, 1) \quad (1)$$

If there are heteroskedasticities or serial correlations, (1) does not hold because we do not have $\text{Var}(\hat{\beta}|X) = \sigma^2(X'X)^{-1}$ and thus cannot estimate $\widehat{\text{Var}(\hat{\beta}|X)}$ using $\widehat{\text{Var}(\hat{\beta}|X)} = \hat{\sigma}^2(X'X)^{-1}$. There are two remedies.

1. (GLS): Transform the original data so that the error terms are nicely behaved.
2. (HAC): Use an estimator for $\text{Var}(\hat{\beta}|X)$ that is robust to heteroskedasticity and serial correlation.

2.1 GLS

The problem is that we have

$$\text{Var}(U|X) \neq \sigma^2 I_n$$

To remedy this, we use a transformation U^* of U with the property $\text{Var}(U^*|X) = I_n$. Let

$$\Sigma^{-1} = \Gamma\Gamma'$$

so that we have

$$\Sigma = \Gamma'^{-1}\Gamma^{-1}$$

Then, we have

$$\Gamma'Y = \Gamma'X\beta + \Gamma'U$$

Define Y^*, X^*, U^* so that

$$Y^* = X^*\beta + U^*$$

The idea is to do OLS using the above expression. Note that we have

$$\begin{aligned} \text{Var}(U^*|X^*) &= \text{Var}(U^*|X) \\ &= \text{Var}(\Gamma'U|X) \\ &= \Gamma'\text{Var}(U|X)\Gamma \\ &= \Gamma'\Sigma\Gamma \\ &= \Gamma'\Gamma'^{-1}\Gamma^{-1}\Gamma \\ &= I_n \end{aligned}$$

GLS is BLUE because the regression $Y^* = X^*\beta + U^*$ does not suffer from heteroscedasticities or serial correlation. The GLS estimator is given by

$$\begin{aligned}\hat{\beta}_{GLS} &= (X^{*'}X^*)^{-1}X^{*'}Y^* \\ &= (X'\Gamma\Gamma'X)^{-1}X'\Gamma\Gamma'Y \\ &= (X'\Sigma^{-1}X)^{-1}X'\Sigma^{-1}Y\end{aligned}$$

The variance of the GLS estimator is given by

$$\begin{aligned}\text{Var}(\hat{\beta}_{GLS}|X) &= \text{Var}((X'\Sigma^{-1}X)^{-1}X'\Sigma^{-1}U|X) \\ &= (X'\Sigma^{-1}X)^{-1}X'\Sigma^{-1}\text{Var}(U|X)\Sigma^{-1}X(X'\Sigma^{-1}X)^{-1} \\ &= (X'\Sigma^{-1}X)^{-1}\end{aligned}$$

2.2 HAC

Instead of GLS, we may find a consistent estimator for $\Omega = \frac{1}{n}X'\Sigma X$. Let this estimator be given by $\hat{\Omega}$. Then, we may construct

$$\widehat{\text{Var}(\hat{\beta}|X)} = n(X'X)^{-1}\hat{\Omega}(X'X)^{-1}$$

How do we estimate Ω ? Refer to Newey-West (1987).

3 The Riemann-Stieltjes Integral

¹ **Definitions** Let $[a, b]$ be a given interval.

1. A **partition** P of $[a, b]$ is a finite set of points $P = \{x_0, x_1, \dots, x_n\}$ such that

$$a = x_0 < x_1 < \dots < x_{n-1} < x_n = b$$

We write $\Delta x_i = x_i - x_{i-1}$ for $i = 1, 2, \dots, n$ and define the mesh π_n as

$$\pi_n = \max_{1 \leq i \leq n} |x_i - x_{i-1}|$$

2. Let α be a monotonically increasing function on $[a, b]$, f a bounded function on $[a, b]$. Corresponding to each partition P of $[a, b]$, we define

$$U(P, f, \alpha) = \sum_{j=1}^n M_j(f) \Delta \alpha_j, \quad L(P, f, \alpha) = \sum_{j=1}^n m_j(f) \Delta \alpha_j$$

as the **upper and lower Riemann-Stieltjes sum** where

$$M_j(f) = \sup_{x \in [x_{j-1}, x_j]} f(x), \quad m_j(f) = \inf_{x \in [x_{j-1}, x_j]} f(x)$$

and

$$\Delta \alpha_j = \alpha(x_j) - \alpha(x_{j-1})$$

for $j = 1, \dots, n$.

¹Please refer to "Principles of Mathematical Analysis" by Rudin for more details.

3. We define the **upper and lower Riemann-Stieltjes integral** as

$$\begin{aligned}\bar{\int}_a^b f d\alpha &= \inf\{U(P, f, \alpha) | P \text{ is partition of } [a, b]\} \\ \underline{\int}_a^b f d\alpha &= \sup\{L(P, f, \alpha) | P \text{ is partition of } [a, b]\}\end{aligned}$$

If f is bounded so that there exists m, M such that $m \leq f(x) \leq M$ for all $x \in [a, b]$, we have

$$m \leq m_j(f) \leq M_j(f) \leq M$$

for all $j = 1, \dots, n$ and hence

$$m(\alpha(b) - \alpha(a)) \leq L(P, f, \alpha) \leq U(P, f, \alpha) \leq M(\alpha(b) - \alpha(a))$$

Thus, the upper and lower Riemann-Stieltjes integral exists for every bounded function f .

4. We say that f is **Riemann-Stieltjes integrable** on $[a, b]$ with respect to α if we have

$$\bar{\int}_a^b f d\alpha = \underline{\int}_a^b f d\alpha$$

In this case, we write

$$\int_a^b f d\alpha := \bar{\int}_a^b f d\alpha = \underline{\int}_a^b f d\alpha$$

A Riemann-Stieltjes integrable function f on $[a, b]$ with respect to the identity function $\alpha(x) = x$ is called **Riemann integrable**.

Theorem 2. *A bounded function f is Riemann-Stieltjes integrable on $[a, b]$ if and only if for each $\epsilon > 0$, there exists a partition P of $[a, b]$ such that*

$$U(P, f, \alpha) - L(P, f, \alpha) < \epsilon$$

Theorem 3. *If a bounded function f is Riemann-Stieltjes integrable on $[a, b]$, there exists a partition $P = \{x_0, x_1, \dots, x_n\}$ of $[a, b]$ such that*

$$\left| \sum_{i=1}^n f(t_i) \Delta \alpha_i - \int_a^b f d\alpha \right| < \epsilon$$

for any arbitrary points $t_i \in [x_{i-1}, x_i]$ for all $i = 1, \dots, n$.

Theorem 4. *Let f be a bounded function on $[a, b]$. Suppose f has only finitely many points of discontinuity on $[a, b]$ and that α is continuous at every point at which f is discontinuous. Then f is Riemann-Stieltjes integrable on $[a, b]$.*

The proofs of the theorems above are given in p.124-p.127 of Rudin (1976).
We define the total variation of α on $[a, b]$ as

$$V(\alpha, [a, b]) = \sup \left\{ \sum_{i=1}^n |\alpha(x_i) - \alpha(x_{i-1})| : P = \{x_0, \dots, x_n\} \text{ is a partition of } [a, b] \right\}$$

Recall that in all of the discussions above, we've assumed that $\alpha : [a, b] \rightarrow \mathbb{R}$ is a monotonically increasing function. However, we may instead assume that $\alpha : [a, b] \rightarrow \mathbb{R}$ is of bounded variation, i.e., $V(\alpha, [a, b]) < \infty$. This is because a function is of bounded variation if and only if it is the difference between two (bounded) increasing functions. To see this, note that if α is of bounded variation, the function $\alpha_1 : [a, b] \rightarrow \mathbb{R}$ defined as

$$\alpha_1(x) = \begin{cases} V(\alpha, [a, x]) & x \in (a, b] \\ 0 & x = a \end{cases}$$

is an increasing function. Also define $\alpha_2 : [a, b] \rightarrow \mathbb{R}$ as

$$\alpha_2(x) = \alpha_1(x) - \alpha(x)$$

Then, we see that for any x, y such that $a \leq x < y \leq b$, we have

$$\begin{aligned} \alpha_1(y) - \alpha_1(x) &= V(\alpha, [y, x]) \\ &\geq \alpha(y) - \alpha(x) \end{aligned}$$

which implies

$$\alpha_2(y) \geq \alpha_2(x)$$

Thus, we may write $\alpha = \alpha_1 - \alpha_2$ where both α_1 and α_2 are increasing functions.

Next, we prove the fundamental theorem of calculus for the Riemann integral.

Theorem 5 (FTC). *Let f be a Riemann integrable function on $[a, b]$. If there is a differentiable function F on $[a, b]$ such that*

$$F'(x) = f(x)$$

for all $x \in [a, b]$, we have

$$F(b) - F(a) = \int_a^b f(x) dx$$

Proof. Let $\epsilon > 0$ be given. Choose a partition $P = \{x_0, \dots, x_n\}$ of $[a, b]$ satisfying the condition given in Theorem 3, i.e.,

$$\left| \sum_{i=1}^n f(t_i) \Delta x_i - \int_a^b f(x) dx \right| < \epsilon$$

for any arbitrary points $t_i \in [x_{i-1}, x_i], i = 1, \dots, n$. Note that due to the mean value theorem, we may write

$$\begin{aligned} F(b) - F(a) &= \sum_{i=1}^n [F(x_i) - F(x_{i-1})] \\ &= \sum_{i=1}^n F'(t_i) \Delta x_i \end{aligned}$$

for some $t_i \in (x_{i-1}, x_i)$ for all $i = 1, \dots, n$. Using $F'(t_i) = f(t_i)$, we see that

$$\left| F(b) - F(a) - \int_a^b f(x)dx \right| < \epsilon$$

Since this holds for any $\epsilon > 0$, the proof is complete. $\square$

Theorem 6 (Integration by Parts). *Suppose F and G are differentiable functions on $[a, b]$, and both $F' = f$ and $G' = g$ are Riemann integrable. Then,*

$$\int_a^b F(x)g(x)dx = F(b)G(b) - F(a)G(a) - \int_a^b f(x)G(x)dx$$

Proof. Write $H(x) = F(x)G(x)$ and apply the fundamental theorem of calculus to H and H' . $\square$

Note that in writing the Riemann integral

$$\int_a^b f(x)dx$$

x is a variable of integration and plays no essential role in the definition of the integral. We may as well write $\int_a^b f(y)dy$ or $\int_a^b f(t)dt$ with no confusion as to what they mean. Furthermore, as long as $g : [a, b] \rightarrow \mathbb{R}$ is continuous and strictly increasing, we have

$$\int_a^b f(x)dx = \int_A^B f(g(y))dg(y) \quad (2)$$

where $A = g^{-1}(a)$ and $B = g^{-1}(b)$ by the change of variables formula. Using the fundamental theorem of calculus on $\int_a^b f(x)dx$ and taking notice of the equality in (2), we may write

$$\int_A^B f(g(y))dg(y) = F(g(B)) - F(g(A)) \quad (3)$$

in particular since $F(g(A)) = F(a)$ and $F(g(B)) = F(b)$. Although we have derived Equation (3) starting with g continuous and strictly increasing, we may relax this condition to g continuous and of bounded variation by regarding

$$\int_A^B f(g(y))dg(y) = \int_A^B h dg \quad (4)$$

as a Riemann-Stieltjes integral of the function $h : [A, B] \rightarrow \mathbb{R}$ defined as $h(y) = f(g(y))$ with respect to $g : [A, B] \rightarrow \mathbb{R}$. To see this more clearly, we present and prove a generalized fundamental theorem of calculus, as shown below.

Theorem 7 (GFTC). *Let $g : [A, B] \rightarrow \mathbb{R}$ be of bounded variation. Let $h : [A, B] \rightarrow \mathbb{R}$ defined as $h(y) = f(g(y))$ be a Riemann-Stieltjes integrable function with respect to g . Suppose there is a differentiable function F on $[a, b]$ such that*

$$F'(x) = f(x)$$

for all $x \in [a, b]$, where $a = g(A)$ and $b = g(B)$. Then, we have

$$F(g(B)) - F(g(A)) = \int_A^B f(g(y))dg(y)$$

Proof. Let $\epsilon > 0$ be given. Choose a partition $P = \{y_0, \dots, y_n\}$ of $[A, B]$ satisfying the condition given in Theorem 3, i.e.,

$$\left| \sum_{i=1}^n f(g(t_i)) \Delta g_i - \int_A^B f(g(y)) dg(y) \right| < \epsilon$$

for any arbitrary points $t_i \in [y_{i-1}, y_i], i = 1, \dots, n$. Note that by the mean value theorem, we may write

$$\begin{aligned} F(g(B)) - F(g(A)) &= \sum_{i=1}^n [F(g(y_i)) - F(g(y_{i-1}))] \\ &= \sum_{i=1}^n F'(\tilde{g}_i) \Delta g_i \end{aligned}$$

for some $\tilde{g}_i$ on the line connecting $g(y_i)$ and $g(y_{i-1})$ for all $i = 1, \dots, n$. Since g is continuous, we may find $t_i \in [y_{i-1}, y_i]$ such that $g(t_i) = \tilde{g}_i$ for all i . Thus, we may write

$$F(g(B)) - F(g(A)) = \sum_{i=1}^n F'(g(t_i)) \Delta g_i$$

for some $t_i \in [y_{i-1}, y_i]$. Using $F'(x) = f(x)$, we see that

$$\left| F(g(B)) - F(g(A)) - \int_A^B f(g(y)) dg(y) \right| < \epsilon$$

Since this holds for any $\epsilon > 0$, the proof is complete. $\square$

Thus, we obtain a generalized fundamental theorem of calculus in the case where $h(y) = h(g(y))$ is Riemann-Stieltjes integrable w.r.t. g and g is continuous and of bounded variation.

4 Stochastic Calculus

4.1 Riemann integral of Stochastic Process

Riemann integral of a Brownian motion is well defined as a random variable. Consider

$$\int_0^1 W(r) dr \tag{5}$$

We see that for any outcome $\omega \in \Omega$, $W(r)(\omega)$ is continuous w.r.t. r and that r is of bounded variation. Thus, the Riemann integral exists for each $\omega \in \Omega$. If we define (5) as a function that maps an outcome $\omega \in \Omega$ to the value $\int_0^1 W(r)(\omega) dr$, it is a well-defined random variable.

But what is the random variable represented by $\int_0^1 W(r)dr$? Notice that we have

$$\begin{aligned}\int_0^1 W(r)dr &= \lim_{\pi_n \rightarrow 0} \sum_{i=1}^n W(s_i)(t_i - t_{i-1}) \\ &= \lim_{\pi_n \rightarrow 0} \sum_{i=1}^n W(t_{i-1})(t_i - t_{i-1}) \\ &= \lim_{\pi_n \rightarrow 0} \left(\sum_{i=1}^n W(t_{i-1}) \right) \frac{1}{n}\end{aligned}$$

where the first equality holds for any choice of partition (t_i) and any $s_i \in [t_{i-1}, t_i]$, $i = 1, \dots, n$. We just fix $s_i = t_{i-1}$ in the second equality. For the third equality, we use (t_i) such that $t_i - t_{i-1} = \frac{1}{n}$ for all $i = 1, \dots, n$. Now, we introduce a new random variable $Z_i := W(t_i) - W(t_{i-1})$. We see that Z_i is distributed as

$$Z_i \sim_d \mathbb{N}(0, t_i - t_{i-1}) = \mathbb{N}\left(0, \frac{1}{n}\right)$$

and that $Z_i \perp Z_j$ for $i \neq j$ due to orthogonal increment property of $W(r)$. Thus, we may write

$$\begin{aligned}\int_0^1 W(r)dr &= \lim_{\pi_n \rightarrow 0} \left(\sum_{i=1}^n W(t_{i-1}) \right) \frac{1}{n} \\ &= \lim_{\pi_n \rightarrow 0} (Z_{n-1} + 2Z_{n-2} + \dots + nZ_0) \frac{1}{n} \\ &= \lim_{\pi_n \rightarrow 0} \mathbb{N}\left(0, \frac{n(n+1)(2n+1)}{6n}\right) \frac{1}{n} \\ &= \lim_{\pi_n \rightarrow 0} \mathbb{N}\left(0, \frac{n(n+1)(2n+1)}{6n^3}\right) \\ &= \mathbb{N}\left(0, \frac{1}{3}\right)\end{aligned}$$

We see that $\int_0^1 W(r)dr$ is a normal random variable with variance $1/3$.

4.2 Ito Integral

Defining the integral of the form

$$\int_0^t f(r)dB(r) \quad \text{or} \quad \int_0^1 B(r)dB(r)$$

is much trickier. These are not well defined as a path-by-path Riemann-Stieltjes integral because, for any $\omega \in \Omega$, $B(r)(\omega)$ is of unbounded variation. For generality, consider a continuous stochastic process $K = (K(t))$ and a continuous martingale $M = (M(t))$. We define Ito integral of K with respect to M as

$$\int_0^t K(s)dM(s) = \text{plim}_{\pi_n \rightarrow 0} \sum_{i=1}^n K(t_{i-1})(M(t_i) - M(t_{i-1})) \quad (6)$$

where $0 = t_0 < t_1 < \dots < t_n = t$ is a partition of $[0, t]$. For the Ito integral to exist, Equation (6) should be uniquely defined regardless of the choice of the partition. On this note, we will not worry too much about the existence of the Ito integral and assume that the integral exists.

4.3 Total and Quadratic Variations

1. For a stochastic process $X = (X(t))$, we define

$$V(t) = \lim_{\pi_n \rightarrow 0} \sum_{i=1}^n |X(t_i) - X(t_{i-1})|$$

as the total variation of X .

2. For a stochastic process $X = (X(t))$, we define

$$[X](t) = \text{plim}_{\pi_n \rightarrow 0} \sum_{i=1}^n (X(t_i) - X(t_{i-1}))^2$$

as the quadratic variation of X .

Note that both the total variation and quadratic variations, whenever they exist, are also stochastic processes.

4.4 Ito Formula

We are interested in characterizing an integral of the form

$$\int_0^t f(M(s)) dM(s)$$

Recall the generalized fundamental theorem of calculus.

$$F(g(B)) - F(g(A)) = \int_A^B f(g(y)) dg(y)$$

We do not have

$$F(M(t)) - F(M(0)) = \int_0^t f(M(s)) dM(s)$$

because $M = (M(t))$ is of unbounded variation. Notice that we have

$$\begin{aligned} F(M(t)) - F(M(0)) &= \sum_{i=1}^n [F(M(t_i)) - F(M(t_{i-1}))] \\ &= \sum_{i=1}^n F'(M(s_i)) [M(t_i) - M(t_{i-1})] \end{aligned}$$

for some $s_i \in [t_{i-1}, t_i]$ because $M(t)$ is continuous. However,

$$\sum_{i=1}^n F'(M(s_i))[M(t_i) - M(t_{i-1})]$$

will not converge to either the Riemann-Stieltjes integral (because $M(t)$ is of unbounded variation) or the Ito integral (because $s_i \neq t_{i-1}$). To remedy this, we may consider the second order Taylor approximation of $F(M(t_i))$ at $F(M(t_{i-1}))$

$$\begin{aligned} F(M(t)) - F(M(0)) &= \sum_{i=1}^n [F(M(t_i)) - F(M(t_{i-1}))] \\ &= \sum_{i=1}^n F'(M(t_{i-1}))[M(t_i) - M(t_{i-1})] + \frac{1}{2} \sum_{i=1}^n F''(M(s_i))[M(t_i) - M(t_{i-1})]^2 \end{aligned}$$

where the second equality holds for some $s_i \in [t_{i-1}, t_i]$ because $M(t)$ is continuous. Now, we have

$$\sum_{i=1}^n F'(M(t_{i-1}))[M(t_i) - M(t_{i-1})] \rightarrow_p \int_0^t f(M(s))dM(s)$$

by the definition of the Ito integral and

$$\frac{1}{2} \sum_{i=1}^n F''(M(s_i))[M(t_i) - M(t_{i-1})]^2 \rightarrow_{a.s.} \int_0^t F''(M(s))d[M](s)$$

because the partition $P = \{t_0, \dots, t_n\}$ defines $\Delta[M]_i = [M](t_i) - [M](t_{i-1}) \approx [M(t_i) - M(t_{i-1})]^2$. Notice that $[M](t)$ is of bounded variation and $\int_0^t F''(M(s))d[M](s)$ is a well-defined Riemann-Stieltjes integral, which is unique regardless of the choice of $s_i \in [t_{i-1}, t_i]$. Thus, we obtain the Ito formula given as

$$f(M(t)) - f(M(0)) = \int_0^t f'(M(s))dM(s) + \frac{1}{2} \int_0^t f''(M(s))d[M](s)$$

Symbolically, we may write

$$df(M_t) = f'(M_t)dM_t + \frac{1}{2}f''(M_t)d[M]_t$$

by writing $M(t) = M_t$.

4.5 Ito's Isometry

We have

$$\mathbb{E} \left[\left(\int_0^t X_s dW_s \right)^2 \right] = \mathbb{E} \left(\int_0^t X_s^2 ds \right)$$

where W_t is standard Brownian motion and X_t is an adapted process with respect to the natural filtration $\mathcal{F}_t = \sigma((W_u)_{u \leq t})$. This is useful in computing variances of random variables that are given as Ito integrals.

Proof. We first express the Ito integral as a partial sum:

$$\begin{aligned}\int_0^t X_s dW_s &= \lim_{\pi_n \rightarrow 0} \sum_{i=1}^n X_{t_{i-1}} (W_{t_i} - W_{t_{i-1}}) \\ \Rightarrow \left(\int_0^t X_s dW_s \right)^2 &= \lim_{\pi_n \rightarrow 0} \left(\sum_{i=1}^n X_{t_{i-1}} (W_{t_i} - W_{t_{i-1}}) \right)^2\end{aligned}$$

Note that

$$\begin{aligned}\mathbb{E} \left[\left(\sum_{i=1}^n X_{t_{i-1}} (W_{t_i} - W_{t_{i-1}}) \right)^2 \right] \\ = \sum_{i=1}^n \mathbb{E} \left[X_{t_{i-1}}^2 (W_{t_i} - W_{t_{i-1}})^2 \right] + 2 \sum_{j>i} \mathbb{E} \left[X_{t_{i-1}} X_{t_{j-1}} (W_{t_i} - W_{t_{i-1}}) (W_{t_j} - W_{t_{j-1}}) \right]\end{aligned}$$

Here,

$$\begin{aligned}\sum_{i=1}^n \mathbb{E} \left[X_{t_{i-1}}^2 (W_{t_i} - W_{t_{i-1}})^2 \right] &= \sum_{i=1}^n \mathbb{E} \left[\mathbb{E} \left[X_{t_{i-1}}^2 (W_{t_i} - W_{t_{i-1}})^2 | \mathcal{F}_{t_{i-1}} \right] \right] \\ &= \sum_{i=1}^n \mathbb{E} \left[X_{t_{i-1}}^2 \mathbb{E} \left[(W_{t_i} - W_{t_{i-1}})^2 | \mathcal{F}_{t_{i-1}} \right] \right] \\ &= \sum_{i=1}^n \mathbb{E} \left[X_{t_{i-1}}^2 \mathbb{E} \left[(W_{t_i} - W_{t_{i-1}})^2 \right] \right] \\ &= \sum_{i=1}^n \mathbb{E} \left[X_{t_{i-1}}^2 (t_i - t_{i-1}) \right] \\ &\rightarrow \mathbb{E} \left[\int_0^t X_s^2 ds \right]\end{aligned}$$

as $\pi_n \rightarrow 0$ where we've use the law of iterated expectation, adaptedness of X_t , independent increment of W_t , and variance of W_t in each equalities. Also,

$$\begin{aligned}\sum_{j>i} \mathbb{E} \left[X_{t_{i-1}} X_{t_{j-1}} (W_{t_i} - W_{t_{i-1}}) (W_{t_j} - W_{t_{j-1}}) \right] &= \sum_{j>i} \mathbb{E} \left[\mathbb{E} \left[X_{t_{i-1}} X_{t_{j-1}} (W_{t_i} - W_{t_{i-1}}) (W_{t_j} - W_{t_{j-1}}) | \mathcal{F}_{t_{j-1}} \right] \right] \\ &= \sum_{j>i} \mathbb{E} \left[X_{t_{i-1}} X_{t_{j-1}} (W_{t_i} - W_{t_{i-1}}) \mathbb{E} \left[(W_{t_j} - W_{t_{j-1}}) | \mathcal{F}_{t_{j-1}} \right] \right] \\ &= \sum_{j>i} \mathbb{E} \left[X_{t_{i-1}} X_{t_{j-1}} (W_{t_i} - W_{t_{i-1}}) \mathbb{E} \left[(W_{t_j} - W_{t_{j-1}}) \right] \right] \\ &= 0\end{aligned}$$

where we've used the law of iterated expectation, adaptedness of X_t and W_t , independent increment of W_t , and zero mean of W_t in each inequality. Thus,

$$\begin{aligned}\mathbb{E} \left(\int_0^t X_s dW_s \right)^2 &= \lim_{\pi_n \rightarrow 0} \mathbb{E} \left(\sum_{i=1}^n X_{t_{i-1}} (W_{t_i} - W_{t_{i-1}}) \right)^2 \\ &= \mathbb{E} \left(\int_0^t X_s^2 ds \right)\end{aligned}$$

which completes the proof. Note that Ito isometry is due to the "conditional uncorrelatedness" of $W_{t_i} - W_{t_{i-1}}$ and $\sigma((W_u)_{u \leq t_{i-1}})$. $\square$

4.6 Examples

1. $\int_0^1 W_s dW_s$
Let $f(W_t) = W_t^2$. Using Ito's formula,

$$d(W_t^2) = 2W_t dW_t + \frac{1}{2}2dt$$

because $[W]_t = t$. Thus, we have

$$\begin{aligned} \int_0^1 1d(W_t^2) &= 2 \int_0^1 W_t dW_t + \int_0^1 1dt \\ W_1^2 &= 2 \int_0^1 W_t dW_t + 1 \\ \int_0^1 W_t dW_t &= \frac{1}{2}(W_1^2 - 1) \end{aligned}$$

2. Consider

$$\int_0^1 f(s) dW_s$$

We know that this is a normal random variable because it is the limit of a weighted sum of increments of Brownian motion. Thus, to find its distribution, we just need to find its mean and variance. The mean is zero. The variance is given by

$$\mathbb{E} \left[\left(\int_0^1 f(s) dW_s \right)^2 \right] = \mathbb{E} \left[\int_0^1 f(s)^2 ds \right] = \int_0^1 f(s)^2 ds$$

where we've used Ito isometry in the first equality. Therefore, we have

$$\int_0^1 f(s) dW_s =_d \mathbb{N} \left(0, \int_0^1 f(s)^2 ds \right)$$

5 Appendix

Stratanovich Integral

Consider again a continuous stochastic process $K = (K(t))$ and a continuous martingale $M = (M(t))$. We define Stratanovich integral of K with respect to M as

$$\int_0^t K(s) \circ dM(s) = \text{plim}_{\pi_n \rightarrow 0} \sum_{i=1}^n \left(\frac{K(t_i) + K(t_{i-1})}{2} \right) (M(t_i) - M(t_{i-1}))$$

where $0 = t_0 < t_1 < \dots < t_n = t$ is a partition of $[0, t]$. We want to prove that the usual fundamental theorem of calculus holds for the Stratanovich integral. In other words,

$$f(M_t) - f(M_0) = \int_0^t f'(M_s) \circ dM_s$$

Proof. As in the proof for the Ito formula, we first expand the difference $f(M_t) - f(M_0)$ by creating a telescoping sum and use second-order Taylor expansion to obtain

$$\begin{aligned} f(M_t) - f(M_0) &= \sum_{i=1}^n [f(M_{t_i}) - f(M_{t_{i-1}})] \\ &= \sum_{i=1}^n f'(M_{t_{i-1}})(M_{t_i} - M_{t_{i-1}}) + \frac{1}{2} \sum_{i=1}^n f''(M_{s_i})(M_{t_i} - M_{t_{i-1}})^2 \end{aligned} \quad (7)$$

for some $s_i \in [t_{i-1}, t_i]$. Now, because the Stratanovich integral is defined with the integrand at $\frac{M_{t_i} + M_{t_{i-1}}}{2}$, we rewrite the first term in (7) as follows:

$$\begin{aligned} &\sum_{i=1}^n f'(M_{t_{i-1}})(M_{t_i} - M_{t_{i-1}}) \\ &= \sum_{i=1}^n f' \left(\frac{M_{t_i} + M_{t_{i-1}}}{2} \right) (M_{t_i} - M_{t_{i-1}}) + \sum_{i=1}^n \left[f'(M_{t_{i-1}}) - f' \left(\frac{M_{t_i} + M_{t_{i-1}}}{2} \right) \right] (M_{t_i} - M_{t_{i-1}}) \end{aligned} \quad (8)$$

The first term in (8) limits to the Stratanovich integral by definition.

$$\lim_{\pi_n \rightarrow 0} \sum_{i=1}^n f' \left(\frac{M_{t_i} + M_{t_{i-1}}}{2} \right) (M_{t_i} - M_{t_{i-1}}) = \int_0^t f'(s) \circ dM_s$$

To characterize the second term of (8), we use the mean value theorem to obtain

$$f'(M_{t_{i-1}}) - f' \left(\frac{M_{t_i} + M_{t_{i-1}}}{2} \right) = f''(M_{s_i^*}) \left(M_{t_{i-1}} - \frac{M_{t_i} + M_{t_{i-1}}}{2} \right)$$

which holds for some $s_i^* \in [t_{i-1}, \frac{t_i + t_{i-1}}{2}]$ by continuity of M_t . Thus, the second term of (8) can be written as

$$\begin{aligned} &\sum_{i=1}^n \left[f'(M_{t_{i-1}}) - f' \left(\frac{M_{t_i} + M_{t_{i-1}}}{2} \right) \right] (M_{t_i} - M_{t_{i-1}}) \\ &= \sum_{i=1}^n f''(M_{s_i^*}) \left(M_{t_{i-1}} - \frac{M_{t_i} + M_{t_{i-1}}}{2} \right) (M_{t_i} - M_{t_{i-1}}) \\ &= -\frac{1}{2} \sum_{i=1}^n f''(M_{s_i^*}) (M_{t_i} - M_{t_{i-1}})^2 \end{aligned} \quad (9)$$

Now we can see that when we take limit as $\pi_n \rightarrow 0$, the $s_i^* \in [t_{i-1}, \frac{t_i + t_{i-1}}{2}]$ and $s_i \in [t_{i-1}, t_i]$ converge to the same point for each $i = 1, \dots, n$ so that (9) cancels out the second term in (7), resulting in usual fundamental theorem of calculus. $\square$